

Post-plant3: The Third Type of a Series of Non-humanoid Robots with an Embedded Physical Interaction

The development of non-verbal human-robot interaction framework and interactive robot prototypes

Keunwook Kim
Seoul National University
kwkim15@snu.ac.kr

ABSTRACT

Post-plant is a series of plant-like robots which communicate non-verbally through physical movements. Most of the social robots look like humans or animals, communicating with us by mimicking human speech and gestures. Inspired by plants, post-plant respond to touch instead of language. With post-plant as a starting point, robots of the future will communicate with us in their own way, without having to mimic human behaviors. Post-plant 3 is the third type of post-plant robot series.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Interaction paradigms;

KEYWORDS

Physical human-robot interaction, Tangible user interface, integrated motor interface

ACM Reference Format:

Keunwook Kim. 2021. Post-plant3: The Third Type of a Series of Non-humanoid Robots with an Embedded Physical Interaction: The development of non-verbal human-robot interaction framework and interactive robot prototypes. In *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21 Adjunct)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3474349.3480206>

1 BACKGROUND

Most of the social robots communicate with voice, gestures, and facial expressions like humans or animals. As they portray the images of existing social creatures, their appearance and interactions imply the social beings behind the robot. Therefore, the violence against those robots can also be violence against humans or animals. Instead of relying on the communication method of humans or animals, there is a design space to build a novel interaction framework based on shape, movements, and other factors including pattern and speed of the motion. Based on the framework, this demo proposes a novel embedded physical interaction system and robot prototypes including the prior research of making the first and second types

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '21 Adjunct, October 10–14, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8655-5/21/10.

<https://doi.org/10.1145/3474349.3480206>



Figure 1: Final robot model, pp-1



Figure 2: Final robot model, pp-2

of post-plant series.[1] Post plant 3 is based on the shape of plants like other series.

2 DESIGNING PP-3(POST PLANT 3)

If portraying images of existing creatures is a problem, what is the difference between mimicking animals and plants? Plants sometimes appear dead because they move much slower than humans. However, they compete, cooperate, live, and die just like other animals. Post plant uses plants as motifs to transform perceptions by representing plants, which are commonly misunderstood as dead objects, and to make a subtle surprise upon the familiar feelings of plants. PP-3(post-plant 3) is based on the shape of seaweed. The robot includes a string and two motors with small tires. The tires pull up the string upwards and as the string is in a loop, it continues to rotate vertically against the gravity. This makes users feel like PP-3 is alive, as most of the test users reacted to the dynamic movements of the string. They first hesitated to touch the string, but once they touched it, they found it miraculous, threatening, or entertaining.

3 CONCLUSION

This demo proposes the third type of a non-humanoid robot series. It provide a possibility that humans and robots can communicate in a wider variety of ways beyond the existing language and



Figure 3: Final robot model, pp-3

gesture-based interactions through a novel human-robot interaction method. This enables communication with more diverse forms and functions of robots in various ways. This prototype can be a primary example for the development of robots using non-verbal interactions in the future.

ACKNOWLEDGMENTS

The background research was funded by Undergraduate Research Program and Student Directed Education course from Faculty of

Liberal Education, Seoul National University. The motor interface system of post plant 1 and 2 is engineered by Jaeyeon Park and Taeyun Li.

REFERENCES

- [1] Kim, K., Park, J., & Li, T. (2021, May). Post-plant: A Series of Non-humanoid Robots with Embedded Physical Non-verbal Interaction: The Development of Non-Verbal Human-Robot Interaction Framework and Input/Output Integrated Motor Interface. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (pp. 1-3).